

QUANTUM BASICS • BEGINNER NOTES

Sections 01–02

GUIDED LESSON • LEARN BY EXPERIMENTING

Information & Qubits

A from-scratch introduction to quantum information: begin with the simplest possible choice, understand the classical bit, and then discover what makes a qubit different.

01 Information Starts With a Choice

Before quantum, understand the language of information

1. What does “information” actually mean?

Information is anything that helps us **distinguish between different possibilities**.

Imagine asking a simple question: “Is the light switched on?” There are two possible answers — **yes** or **no**.



2. Every choice can be represented

Computers need a way to store these choices. The simplest way is to give each possibility a symbol. For example, we can decide that **0 means NO** and **1 means YES**. Now the answer can be stored using just one binary digit.

EXAMPLE

Question: **Is the door open?**

No → **0** Yes → **1**

3. Meet the bit

A **bit** is the basic unit of classical information. The word “bit” comes from *binary digit*. A bit can have only two possible values: **0** or **1**.

Bit = 0

The bit is in its first possible state. In a simple yes/no system, we could interpret this as “No”.

Bit = 1

The bit is in its second possible state. In the same system, we could interpret this as “Yes”.

4. Why do computers use 0 and 1?

Physical devices can distinguish between different electrical conditions. These conditions can be represented using the labels **0** and **1**. Large collections of these simple choices allow computers to store and process information.

THINK OF IT THIS WAY

One bit gives you **2 possibilities**. Two bits give you **4 possibilities**. Three bits give you **8 possibilities**. In general, **n classical bits can represent 2^n different bit strings**.



5. Bits can be combined

A computer rarely works with just one bit. Bits are grouped together to represent larger pieces of information. A group of eight bits is called a **byte**.

EXAMPLE

The four-bit string **1011** is one possible combination of four classical bits. Since four bits give $2^4 = \mathbf{16}$ **possible strings**, 1011 is simply one of those possibilities.

6. A bit is definite — even if we do not know it

Suppose a coin is hidden underneath a cup. You do not know whether it is heads or tails. But the coin itself is still in one definite state. Classical uncertainty means that **we lack knowledge about the value.**

Unknown is not the same as superposition

If we do not know whether a classical bit is 0 or 1, that is simply lack of information. A quantum superposition is different: the quantum state itself can contain both **|0⟩** and **|1⟩** components.

7. So where does quantum computing begin?

Quantum computing begins with the same basic idea: we need a physical system that can represent information. But instead of using an ordinary bit, we use a **quantum bit — a qubit.**

Classical idea	Meaning
Bit	Basic unit of classical information
Possible values	0 or 1
Several bits	Can form binary strings such as 0101
Unknown bit	The value exists, but we may not know it

KEY IDEA

Classical information starts with a simple choice: **0 or 1**. A bit stores one such choice. Quantum computing keeps these two basic states but introduces a new object — the **qubit** — whose state can behave in ways a classical bit cannot.

02

Meet a Qubit

The smallest unit of quantum information

1. What is a qubit?

A **qubit** is short for **quantum bit**. Just like a classical bit, it has two basic states. We write these states using quantum notation: $|0\rangle$ and $|1\rangle$.

 $|0\rangle$

Read this as “ket zero.” It is one of the two standard basis states of a qubit.

 $|1\rangle$

Read this as “ket one.” It is the second standard basis state of a qubit.

2. What does the $| \rangle$ notation mean?

The vertical bar and angle bracket together are called **ket notation**. It is a mathematical way of writing a quantum state. So when we write $|0\rangle$, we are naming the quantum state “zero”; when we write $|1\rangle$, we are naming the quantum state “one.”

Remember: $|0\rangle$ and $|1\rangle$ are quantum states, not ordinary numbers. They form the basic states used to describe a qubit.

3. A qubit can also be in superposition

This is the first major difference between a classical bit and a qubit. A qubit does not have to be only $|0\rangle$ or only $|1\rangle$. It can exist in a **superposition** of the two states.

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

Here, $|\psi\rangle$ represents the complete state of the qubit. The numbers α and β are called **amplitudes**. They describe the contributions of $|0\rangle$ and $|1\rangle$ to the quantum state.

4. What does “superposition” mean?

Superposition means that the quantum state can be written as a combination of its basis states. It does **not** simply mean that we do not know whether the qubit is 0 or 1.

A simple intuition

Imagine a wave that contains contributions from two different possibilities. The quantum state keeps track of these contributions using amplitudes. Later, these amplitudes can combine and interfere with one another — an idea that becomes very important in quantum computing.

5. The amplitudes contain the important information

The amplitudes α and β are generally complex numbers. We do not directly observe these amplitudes when measuring a qubit. Instead, they determine the probabilities of the possible measurement outcomes.

$$|\alpha|^2 + |\beta|^2 = 1$$

This is called the **normalization condition**. It means that all possible measurement probabilities must add up to 1, or 100%.

6. From amplitude to probability

To find the probability of measuring $|0\rangle$, take the magnitude squared of α . To find the probability of measuring $|1\rangle$, take the magnitude squared of β .

$$P(0) = |\alpha|^2 \quad P(1) = |\beta|^2$$

WORKED EXAMPLE

Suppose a qubit is in the state $0.8|0\rangle + 0.6|1\rangle$.

Probability of 0: $0.8^2 = 0.64 = 64\%$

Probability of 1: $0.6^2 = 0.36 = 36\%$

Therefore: $64\% + 36\% = 100\%$.

7. The famous equal superposition

One of the most important quantum states for beginners is the equal superposition of $|0\rangle$ and $|1\rangle$.

$$|+\rangle = (|0\rangle + |1\rangle) / \sqrt{2}$$

The amplitude of each state is $1/\sqrt{2}$, which is approximately **0.7071**. Squaring this value gives approximately **0.5**. Therefore, each state has a 50% probability of being observed when we measure the qubit.

MEASUREMENT RESULT

For the state $(|0\rangle + |1\rangle)/\sqrt{2}$:

P(0) = 50%

P(1) = 50%

8. Measurement gives one definite result

Before measurement, the qubit can be described by a superposition. When we measure it in the standard computational basis, the result is one definite value: **0 or 1**.

- 1 Prepare the qubit in a quantum state.
- 2 The state contains amplitudes for $|0\rangle$ and $|1\rangle$.
- 3 Measure the qubit.
- 4 One definite result, 0 or 1, is obtained.

9. One measurement does not show the probabilities

Suppose we prepare the state $(|0\rangle + |1\rangle)/\sqrt{2}$. A single measurement might give 0. That does not mean the state had a 100% probability of giving 0. The state still had a 50% chance of producing either result.

IMAGINE REPEATING THE EXPERIMENT

If we prepare the same state many times and measure each qubit, we might see:

0, 1, 1, 0, 0, 1, 0, 1, 1, 0

With many more measurements, the fraction of 0s and 1s approaches approximately **50% each**.

10. How do we create a superposition?

Quantum states can be changed using **quantum gates**. One of the most important gates for beginners is the **Hadamard gate**, usually written as **H**.

FIRST QUANTUM CIRCUIT



11. What does the H gate do?

When the Hadamard gate is applied to $|0\rangle$, it creates an equal superposition of $|0\rangle$ and $|1\rangle$.

$$|0\rangle \rightarrow H \rightarrow (|0\rangle + |1\rangle) / \sqrt{2}$$

Therefore, if we immediately measure the qubit after applying H, there is a **50% chance of obtaining 0** and a **50% chance of obtaining 1**.

12. A qubit is not simply a “better bit”

It is tempting to think of a qubit as a bit that can simply store 0 and 1 at the same time. This is useful as a first intuition, but it is incomplete. The real quantum behaviour comes from the way **amplitudes, phase, superposition and interference** work together.

Classical bit	Qubit
Can be 0 or 1	Has basis states $ 0\rangle$ and $ 1\rangle$
Represents a definite classical value	Can exist in a superposition
Uses ordinary probabilities	Uses quantum amplitudes

Measurement reads the classical value

Measurement produces a probabilistic outcome

No quantum phase

Phase can affect later interference

13. A physical qubit can be many things

A qubit is an abstract mathematical object, but a real quantum computer must represent it using a physical quantum system. Different quantum computers use different physical technologies, including superconducting circuits, trapped ions and photons.

The important part is the quantum behaviour

Whatever physical system is used, we identify two controllable quantum states and label them $|0\rangle$ and $|1\rangle$. The mathematical rules of the qubit then describe how these states can be manipulated and measured.

14. The three ideas to remember

If you are completely new to quantum computing, focus on these three statements first.

- 1 **A qubit has two basis states:** $|0\rangle$ and $|1\rangle$.
 - 2 **A qubit can be in superposition:** $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$.
 - 3 **Measurement gives a definite result:** 0 or 1, with probabilities determined by the amplitudes.
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KEY IDEA

A qubit is the quantum version of a bit. It has two basis states, $|0\rangle$ and $|1\rangle$, but its state can be a superposition of both. The coefficients of the superposition are called amplitudes, and the squared magnitudes of those amplitudes determine the probabilities of measurement outcomes.

QUICK CHECK

1. What are the two basis states of a qubit?
→ $|0\rangle$ and $|1\rangle$

2. What is a superposition?

→ A quantum state written as a combination of basis states.

3. Can we directly measure an amplitude?

→ No. Measurement gives an outcome, while the amplitude determines its probability.

4. What does H do to $|0\rangle$?

→ It creates the equal superposition $(|0\rangle + |1\rangle)/\sqrt{2}$.

5. What can we obtain when we measure that state?

→ Either 0 or 1, each with 50% probability.

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